# The Black-Scholes Model

## 1 The Problem

Before Black, Scholes, and Merton published their work in 1973, there was no rigorous, universally accepted method for pricing options. Practitioners used rules of thumb. The fundamental question of *'what is the fair price of the right to buy a stock at a fixed price on a fixed future date?'* had no clean answer.

The difficulty is that options are non-linear instruments, i.e., the payoff of a call, $\max(S_T - K, 0) - p$, is not proportional to the underlying asset value. Along with this, the fair price today must somehow reflect the full distribution of possible stock prices at expiry, which is itself uncertain. The insight of Black, Scholes, and Merton was to show that, under certain assumptions, you can *eliminate* this uncertainty entirely by constructing a *continuously rebalanced portfolio*. When there is no uncertainty left, the portfolio **must** earn the risk-free rate and that constraint pins down the option price.

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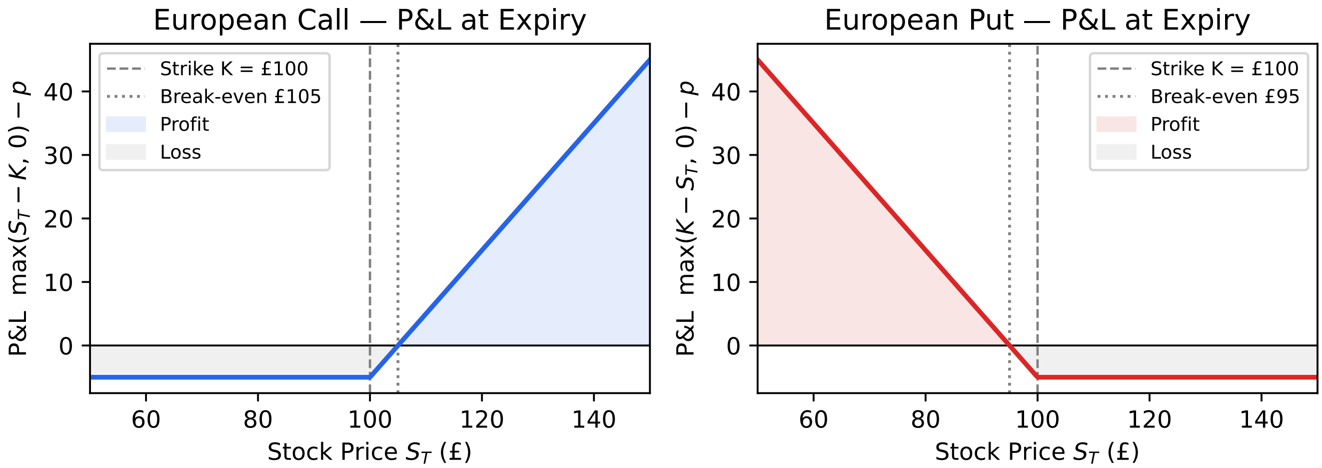


Figure 1: Profit / loss at expiry for a European call and put (K = £100, premium = £5)

## 2 Assumptions

The model rests on several simplifying assumptions. They are not all realistic, but they are tractable, and the model's remarkable robustness has kept it central to options markets for over fifty years.

1. **The underlying follows Geometric Brownian Motion.** Asset prices evolve continuously and randomly, with constant drift and constant volatility.
2. **Volatility is constant.** This is the assumption most obviously violated in practice (hence implied volatility surfaces), but it is the foundation of the analytical solution.
3. **No arbitrage.** There is no riskless profit available in the market.
4. **Frictionless markets.** No transaction costs, no taxes, and the ability to buy or sell any quantity — including fractional amounts — of the underlying continuously.
5. **The risk-free rate is constant.** A fixed rate $r$ is available for borrowing and lending.
6. **No dividends** (in the basic version; the Merton extension handles a continuous dividend yield $q$).

## 3 The Asset Price Model: Geometric Brownian Motion

The starting point is a model for how the underlying asset price $S_t$ evolves over time. Black and Scholes assumed that $S_t$ follows **Geometric Brownian Motion** (GBM):

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t$$

Let's unpack each term:

- $dS_t$ is the instantaneous change in the stock price.
- $\mu$ is the **drift** — the expected rate of return on the stock (e.g. 8% per year). It captures the stock's average upward trend.
- $\sigma$ is the **volatility** — the standard deviation of the stock's log-returns per unit time. This is the single parameter of the Black-Scholes model.
- $dW_t$ is an increment of a **Wiener process** (Brownian motion): a random shock drawn from $\mathcal{N}(0, dt)$ at each instant. These shocks are independent across time.

The $S_t$ in front of $\mu$ and $\sigma$ makes this *geometric*: the uncertainty is proportional to the current price, so the stock can never go negative. This is appropriate for equity prices.

An important consequence of GBM is that **log-returns are normally distributed**. Over a finite horizon $T$, the stock price at expiry satisfies:

$$\ln\left(\frac{S_T}{S_0}\right) \sim \mathcal{N}\left(\left(\mu - \frac{\sigma^2}{2}\right)T,\ \sigma^2 T\right)$$

The $-\frac{\sigma^2}{2}$ term is an Itô correction: because we are taking the log of a process with multiplicative randomness, the mean of the log-return is slightly lower than the drift $\mu$. This is a consequence of Jensen's inequality applied to the concave log function.

Equivalently, the stock price at expiry is **log-normally distributed**:

$$S_T = S_0 \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}\,Z\right], \quad Z \sim \mathcal{N}(0,1)$$

# 4 Risk-Neutral Pricing

Here is the key insight. The option price depends on $\mu$, the expected return on the stock — but $\mu$ is the hardest parameter to estimate and differs across investors. Does this mean every investor arrives at a different option price?

No — and this is Black-Scholes' deepest result. By continuously trading the underlying, you can construct a **replicating portfolio** that perfectly matches the option's payoff, regardless of $\mu$. The portfolio consists of a dynamically-adjusted position in the stock plus a bond. Because it replicates the option exactly, it must have the same price (by no-arbitrage).

The fraction of the portfolio held in the stock — the **hedge ratio** — turns out to be $\frac{\partial V}{\partial S}$, where $V$ is the option value. Applying Itô's lemma to $V(S_t, t)$ and setting up the self-financing replicating portfolio eliminates the $dW_t$ term entirely. The remaining equation must hold with certainty, so it must equal the risk-free rate. This yields the **Black-Scholes PDE**:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

Notice that $\mu$ has completely dropped out. The option price does not depend on how optimistic or pessimistic you are about the stock's expected return.

A cleaner way to arrive at the same result is **risk-neutral pricing**. Because the option price does not depend on $\mu$, we are free to choose any $\mu$ we like without affecting the answer. The most convenient choice is $\mu = r$ — the risk-free rate. Under this *risk-neutral measure* (often denoted $\mathbb{Q}$), all assets earn the risk-free rate in expectation, and the stock price dynamics become:

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t^{\mathbb{Q}}$$

The option price is then simply the **discounted expected payoff under the risk-neutral measure**:

$$V_0 = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\text{Payoff}(S_T)]$$

This is intuitive: price the option as if everyone were risk-neutral (requiring only the risk-free return on all assets), then discount at the risk-free rate. The sleight of hand is that we changed the probability measure, not the actual probabilities — but since the option price doesn't depend on the real-world drift anyway, the answer is correct.

This framework — the risk-neutral measure and the expectation it defines — is exactly what each pricer implements, in different ways. The closed-form pricer evaluates the expectation analytically. The Monte Carlo pricer estimates it by simulation. The binomial tree discretises the process and solves backwards.

## 5 The `BlackScholesModel` Class

`BlackScholesModel` holds the single free parameter of the model: $\sigma$, the constant volatility. Everything else — the current spot price, interest rates, dividends — is observable market data and lives in `MarketData`. The model only encodes what the BS assumption adds: that the asset's randomness is characterised by a single, constant number.

```
from options_pricer import BlackScholesModel

# 20% annualised volatility
model = BlackScholesModel(vol=0.20)

# vol must be strictly positive - zero would degenerate the formula
# BlackScholesModel(vol=0.0)  # raises ValueError
```

The model is passed to a pricer together with an instrument and market data:

```
from options_pricer import (
    BlackScholesModel, EuropeanOption, MarketData, OptionType,
    ClosedFormPricer, MonteCarloPricer, TreePricer,
)

model = BlackScholesModel(vol=0.20)
md    = MarketData(spot=100.0, rate=0.05)
call  = EuropeanOption(option_type=OptionType.CALL, strike=100.0, expiry=1.0)

# All three pricers accept the same (instrument, market_data, model) signature
ClosedFormPricer.price(call, md, model)              # analytical
MonteCarloPricer(n_paths=100_000).price(call, md, model)  # simulation
TreePricer(n_steps=200).price(call, md, model)        # tree
```

## 6 Limitations

No model is complete without a discussion of its shortcomings:

- **Constant volatility** is the most significant failure. In practice, implied volatility varies with strike (the vol smile/skew) and term (the vol term structure). This reflects the market's perception of tail risk and other features not captured by log-normal dynamics. The **Heston model** addresses this by making volatility itself a stochastic process.
- **No jumps.** GBM is a continuous process; in reality, stock prices can gap discontinuously. The **Merton jump-diffusion model** extends BS by superimposing a Poisson jump process.
- **Continuous hedging** is impossible in practice. Discrete hedging introduces error proportional to gamma.
- **Constant interest rates.** For long-dated options, stochastic interest rates matter.

Despite these limitations, Black-Scholes remains the market's shared language. Traders quote implied volatility because it is the number you feed into the BS model to recover the market price. The model may be wrong, but it is *consistently* wrong in ways that practitioners have learned to navigate.